

STRENGTH OF MATERIALS

II Year I Semester: MECHANICAL ENGINEERING								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6ME09	PCC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100
COURSE OVERVIEW: Strength of Materials is the branch of continuum mechanics which deals with behaviour of solids under different loading conditions. This course will help students to acquire knowledge on the concepts needed for the design of various machine elements.								
COURSE OUTCOMES: By undergoing this course, student can be able to: 1. Analyse various stresses, strains and deformations induced in the body due to application of various loads. 2. Evaluate and sketch the variation of shear force and bending moment along the length of different types beams with various support conditions. 3. Evaluate the resistance of various beam sections under bending and shear loads. 4. Analyse the resultant effect of principal stresses induced in the body and can determine slope and deflection for various beams under different loading and boundary conditions. 5. Analyse the stresses developed and deformation of thin and thick cylinder due to internal Pressure.								
UNIT-I	SIMPLE STRESSES AND STRAINS							
Elasticity and Plasticity, Types of stresses and strains, Hooke's law, stress, strain diagram for ductile and brittle materials, Working stress, Factor of safety, Bars of varying section, composite bars, Temperature stresses, Lateral strain, Poisson's ratio and volumetric strain, Elastic module and the relationship between them, Strain energy, Resilience - Gradual, Sudden, and Impact loadings.								
UNIT-II	SHEAR FORCE AND BENDING MOMENT							
Definition of beam, Types of beams, Support reactions, Concept of shear force and bending moment, Relation between Shear Force, Bending Moment and rate of loading at a section of a beam. Shear Force and Bending Moment diagrams for cantilever, simply supported and overhanging beams subjected to point loads, U.D.L., U.V.L and combination of these loads, Point of contra flexure.								
UNIT-III	FLEXURAL STRESSES & SHEAR STRESSES							
FLEXURAL STRESSES: Theory of simple bending, Assumptions, Derivation of bending equation: $M/I = f/y = E/R$, Neutral axis, Moment of Resistance, section modulus and determination bending stresses in rectangular and circular sections (Solid and Hollow), I, T, Angle and Channel sections. SHEAR STRESSES: Derivation of formula, Shear stress distribution across various beams sections like rectangular, circular, I, T, angle sections.								
UNIT-IV	PRINCIPAL STRESSES AND STRAINS & DEFLECTION OF BEAMS							
PRINCIPAL STRESSES AND STRAINS: Introduction - Stresses on an inclined section of a bar under axial loading - compound stresses - Normal and tangential stresses on an inclined plane for biaxial stresses - Two perpendicular normal stresses accompanied by a state of simple shear - Mohr's circle of stresses - Principle stresses and strains - Analytical and graphical solutions.								

DEFLECTION OF BEAMS: Bending into a circular arc slope, deflection and radius of curvature, Differential equation for the elastic line of a beam, Double integration and Macaulay's methods, Determination of slope and deflection for cantilever and simply supported beams subjected to point loads and UDL.

UNIT-V**THIN CYLINDERS & THICK CYLINDERS**

THIN CYLINDERS: Thin seamless cylindrical shells, Derivation of formula for longitudinal and circumferential stresses hoop, longitudinal and volumetric strains, changes in dia, and volume of thin cylinders, Riveted boiler shells, Thin spherical shells.

THICK CYLINDERS: Introduction, Longitudinal Stress, Circumferential Stresses in Thick Cylinders, Lami's Equations, Variation of stresses across the thickness.

Text Books:

1. Dr.S.Sadhu Singh, Strength of materials, Khanna publishers.
2. Ramamrutham S, Strength of materials, Danpath rai publishing company.

Reference Books:

1. Strength of materials by R.K.Bansal, Laxmi Publications (P) Ltd.
2. Engineering mechanics of solids by EGOR P. Popov, Pearson publishers
3. Timoshenko and Gere, Mechanics of Materials, CBS Publishers, 2011
4. Strength of Materials by Andrew Pytel and Ferdinand L. Singer Longman